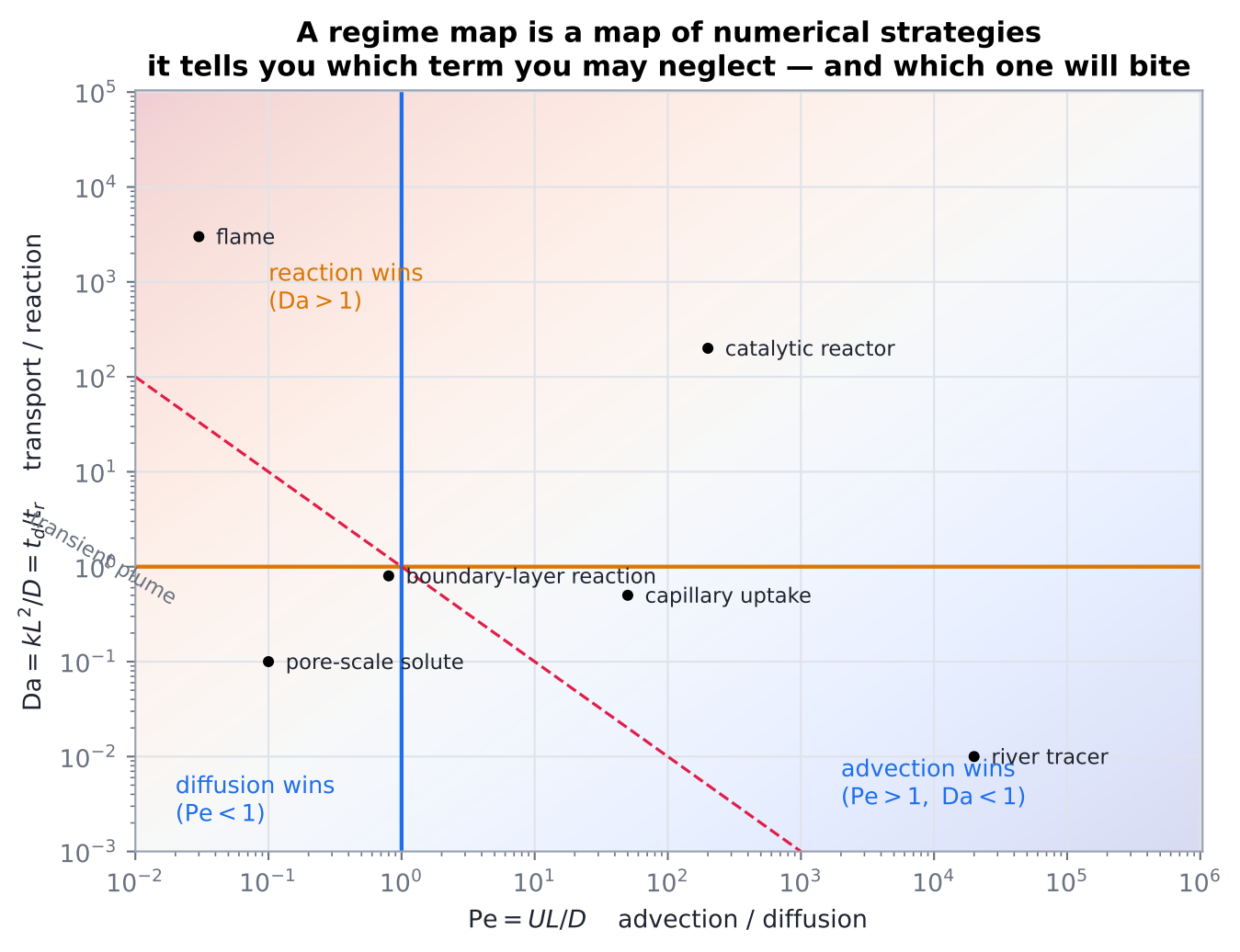
A regime map is a map of numerical strategies
it tells you which term you may neglect — and which one will bite
Da = kL²/D = t_diff/t_r transport / reaction
Pe = UL/D advection / diffusion
flame
reaction wins
(Da > 1)
catalytic reactor
transient plume
boundary-layer reaction
capillary uptake
pore-scale solute
river tracer
diffusion wins
(Pe < 1)
advection wins
(Pe > 1, Da < 1)